

# Biology for Engineers

## Chapter-1: Introduction

The Basics of Life: Chemistry; Organic  
Molecules: The Molecules of Life;

Reference book: Biology for Engineers - Arthur T. Johnson [2nd edition]  
Biology for Engineers – G. K. Suraishkumar

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# Organization of Life:

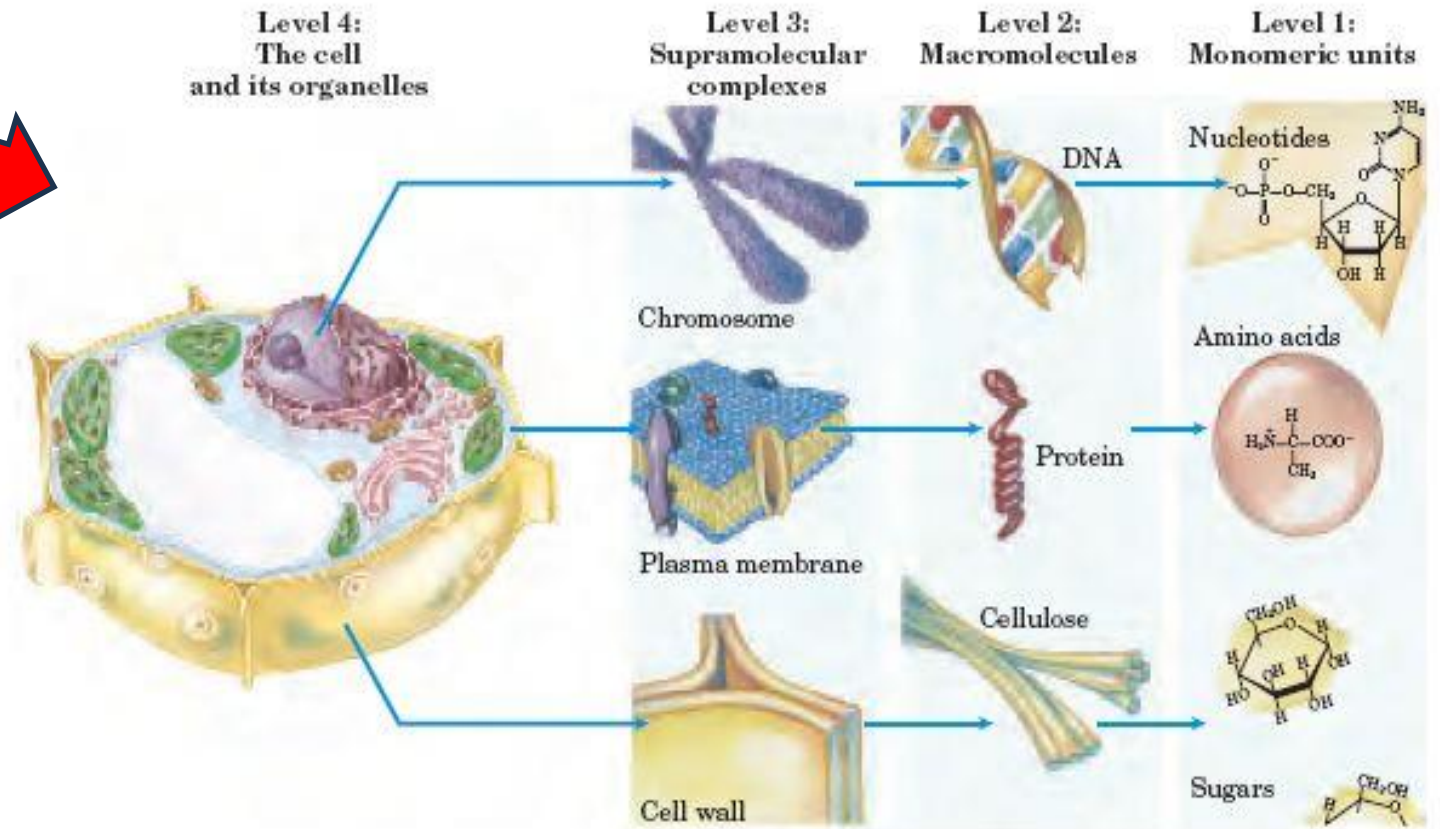
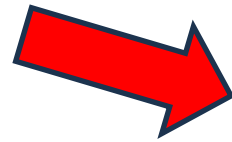
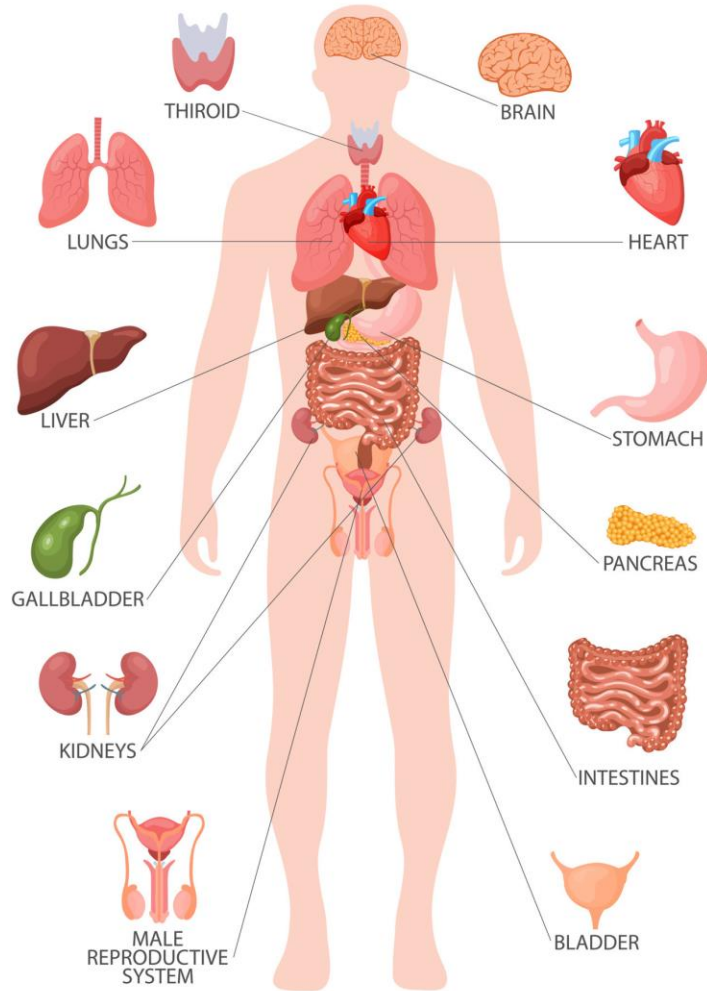
Biomolecule

Cell

Tissue

Organ

Organ system = organism





# History of Biochemistry:

1903, Neuberg (German):

"Biochemistry"

"Chemistry of Life"

# Biochemistry is a branch of life science:

- Which deals with the **Study of Biochemical Reactions and Processes**
- Occurring in living cells of organisms.

"Life" in Biochemistry point of view is:

- **Hundreds of Biochemical reactions** and **Biochemical processes**
- Occurring in **sub cellular organelles** of a cell **in an organized manner.**
- **understanding all biological processes.**
- **diseases in humans, animals and plants**
- **Structure-function, Metabolism, gene regulation etc.**

# Organisms convert food molecules into biomolecules



Some examples of how food molecules are converted into biomolecules:

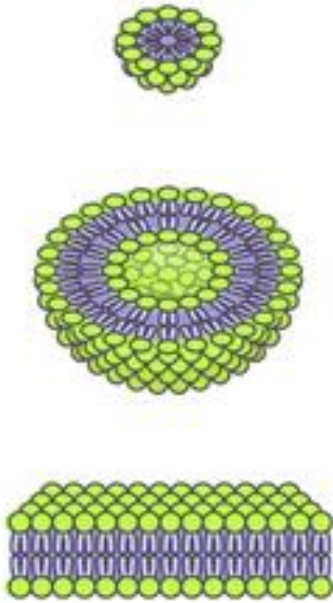
- **Glucose:** Glucose is a monomer that can link with other glucose molecules to form complex carbohydrates like starch.
- **Carbohydrates:** Carbohydrates provide energy, and there are two types: simple and complex.
- **Proteins/Amino Acids:**
- **Fats:** Fats are a food store that living things use extensively.
- **Minerals:** Vitamins and other small minerals



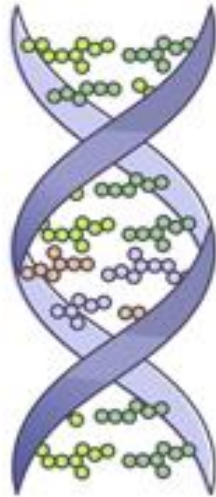
# Types of Biomolecules

There are four main classes of Biomolecules:

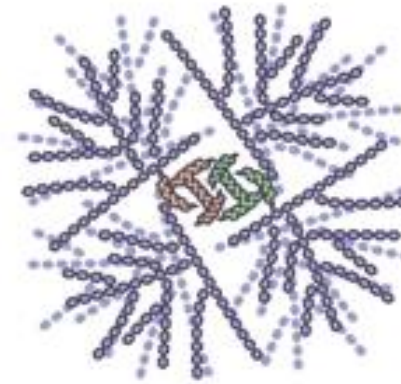
- Carbohydrates
- Proteins
- Nucleic acids
- Lipids



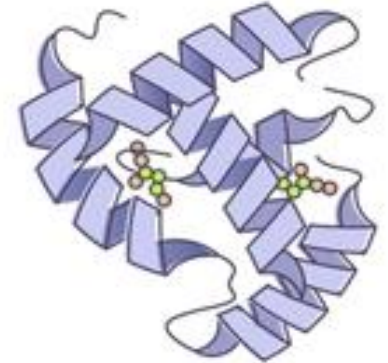
LIPIDS



NUCLEIC ACID

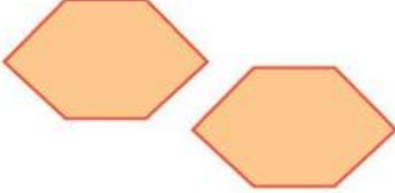
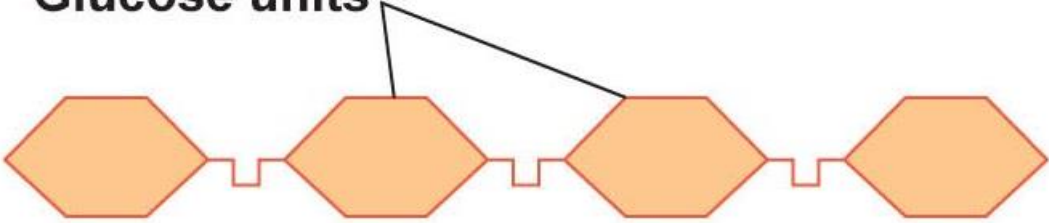
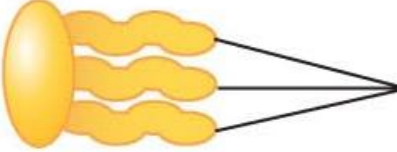
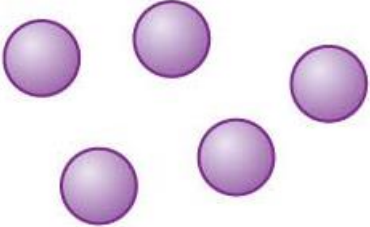
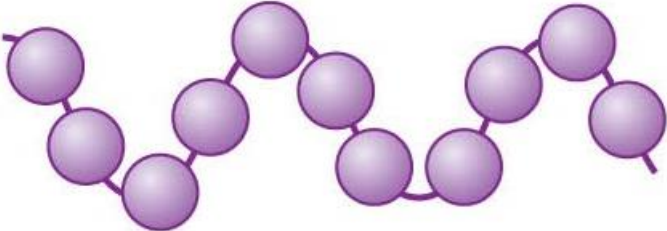


CARBOHYDRATES



PROTEINS

# Biomolecules

| Macronutrients | Chains of                                                                                           | Example                                                                                              |
|----------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Carbohydrates  | Glucose<br>       | Glucose units<br> |
| Lipids         | Fatty acids<br>   | Triglyceride<br>  |
| Proteins       | Amino acids<br> | Amino acids<br> |



# Carbohydrates





# CARBOHYDRATES

- Carbohydrates are the most abundant biomolecules on Earth.
- Each year, photosynthesis converts more than 100 billion metric tons of CO<sub>2</sub> and H<sub>2</sub>O into cellulose and other plant products.

## Function of carbohydrates:

- oxidation of carbohydrates provide most **energy** in **non photosynthetic** cells.
- Insoluble carbohydrate polymers serve as **structural and protective elements** in the cell walls of bacteria and plants and in the connective tissues of animals.
- Other carbohydrates **lubricate skeletal joints** and participate in **recognition and adhesion between cells**.
- More complex carbohydrate polymers covalently attached to **proteins or lipids act as signals** that determine the intracellular location or metabolic fate of these hybrid molecules, called glycoconjugates.



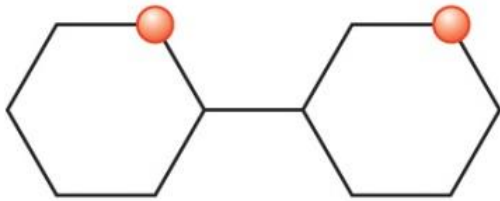
# Classes of Carbohydrates



Monosaccharide

**Carbohydrates:** hydrates of carbon: general formula  $C_n(H_2O)_n$

**Plants:** photosynthesis

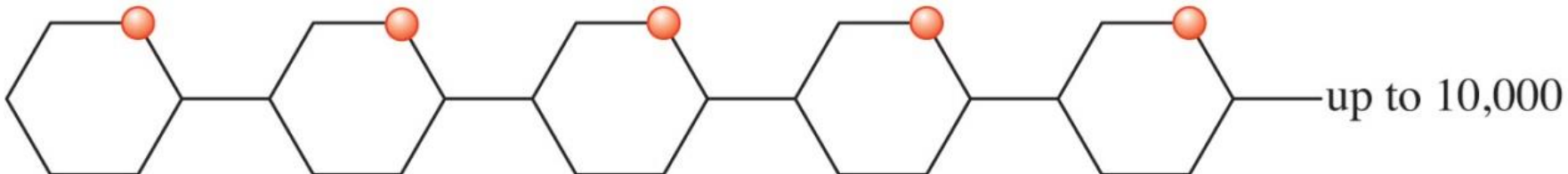


Disaccharide

**Polymers:** large molecules made up of repeating smaller units (monomer)



Oligosaccharide



Polysaccharide

# Classes of Carbohydrates

- **Monosaccharides:** are the simplest carbohydrates. They cannot be broken down to smaller carbohydrates anymore. Ex. **Glucose**
- **Disaccharides:** consist of two monosaccharide units joined together; they can be split into two monosaccharides. **Sucrose** can be broken down into glucose and fructose.
- **Oligosaccharides:** contain anywhere from 3 to 9 monosaccharide units. ABO blood groups are oligosaccharides.
  - Consist of short chain monosaccharide units
  - Joined by glycosidic bonds
  - **Polysaccharides:** are large molecules containing 10 or more monosaccharide units. Carbohydrate units are connected in one continuous chain or the chain can be branched.
  - Contains more than 10 residues
  - Cellulose-linear; Glycogen-branched
  - Ex: Starch, cellulose, chitin

# Complex Carbohydrates: Glycogen –

- Long Chain of glucose found in the animals
- Stored in **liver** and **muscle**
- Helps “**maintain blood glucose**”
- Important source of “**quick energy**”  
especially during fasting, exercise





# Polysaccharides: Starch

**Starch** – long chains of glucose found in plants

1. Cereal grains (wheat, rice, corn, etc.),
2. legumes (beans & peas),
3. root vegetables (Potatoes, Yams, Cassava)

**Potato**



**Cassava**



**Yams**

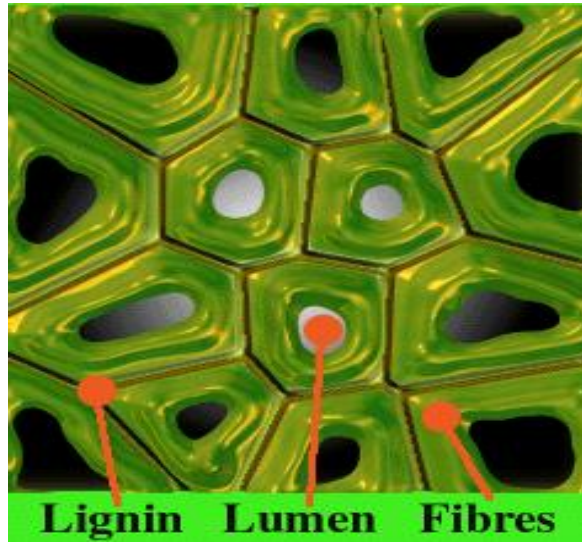


# Polysaccharides: Fiber

**Fiber** – mostly indigestible CHO; gums, mucilages, lignin

1. Component of plant cell walls
2. Classified according to solubility in water
3. Abundant in whole grains, legumes, fruits and vegetables

Types of Legumes



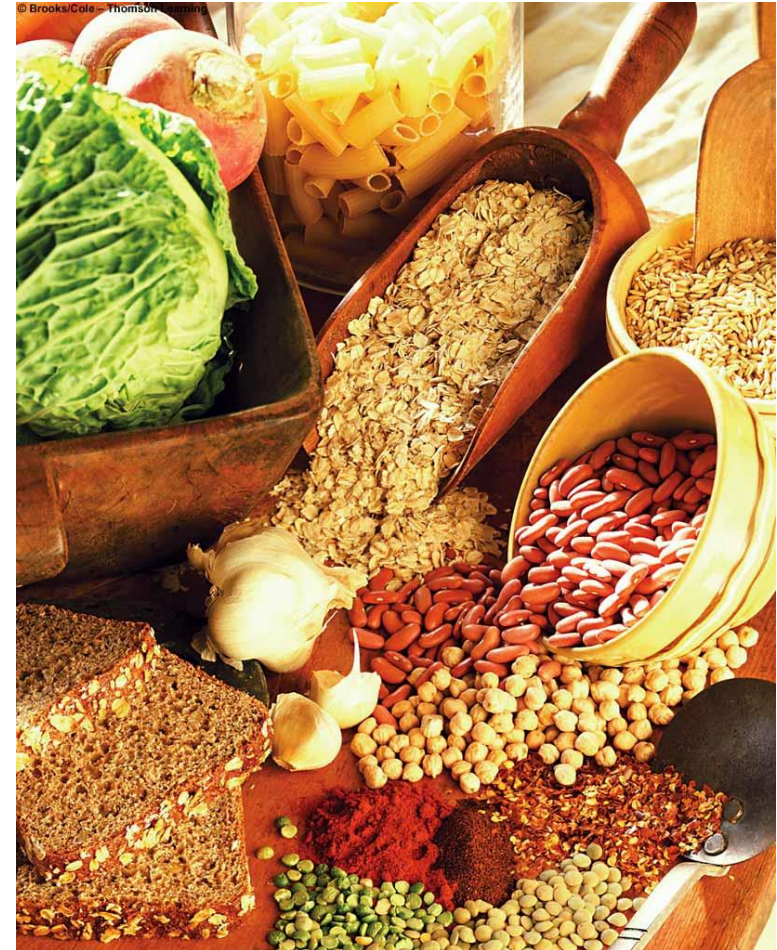
Gums and Mucilages





# Health Benefits of Complex Carbohydrates:

- Complex carbohydrates & fiber may reduce the risk of:
  - Heart disease
  - Diabetes
  - GI health
  - Cancers
  - Weight Management

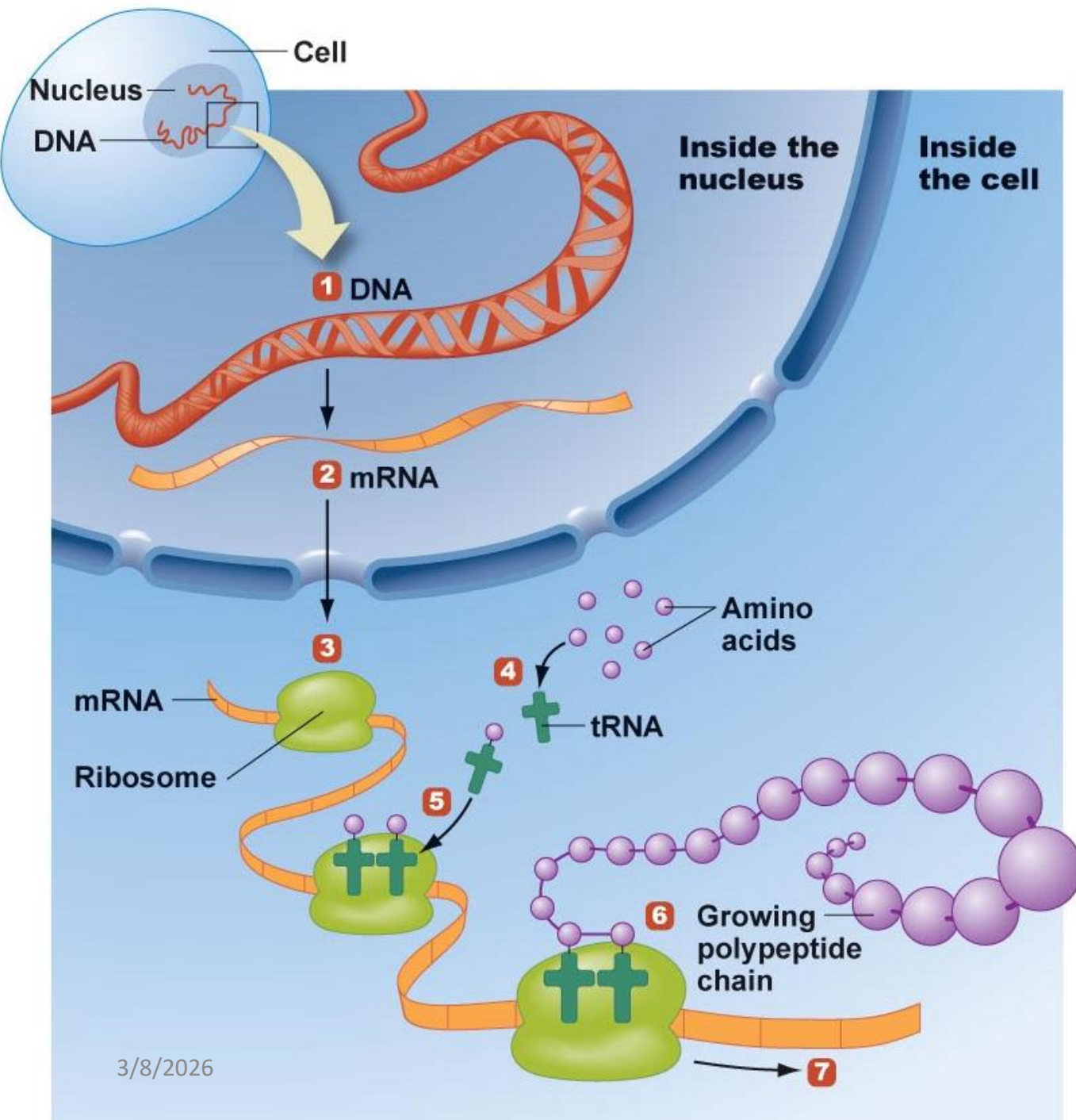






PROTEIN





# What are Proteins?

- Large molecules
- Made up of amino acids chain
- Found in every cell in the body
- Involved in most of the body's functions and life processes
- The sequence of amino acids is determined by DNA

# General functions of proteins

Proteins serve a variety of essential functions in the body. Here are the general roles they play:

**1. Structural Support** – Proteins like collagen, keratin, and elastin provide structural integrity to tissues such as **skin, hair, nails, and connective tissues.**

**1. Enzymatic Activity** – Many proteins function as **enzymes**, which catalyze biochemical reactions necessary for **metabolism, digestion, and cellular processes.**

**2. Transport and Storage** – Proteins like **hemoglobin** transport oxygen, while others help in **transporting nutrients and molecules across cell membranes.**



# General functions of proteins

**4. Cell Signaling and Communication** – Hormonal proteins such as **insulin** regulate bodily functions by transmitting signals between cells.

**5. Immune Defense** – Antibodies (immunoglobulins) are proteins that help the immune system fight infections and pathogens.

**6. Movement and Contraction** – Proteins like **actin** and **myosin** are essential for muscle contraction and movement.

**7. Regulation of Gene Expression** – Some proteins help regulate which genes are turned on or off, influencing cell function and development.

**8. Maintaining Fluid and pH Balance** – Proteins like **albumin** and **globulin** help maintain osmotic balance in the blood and regulate pH levels.

**Each amino acid, aside from its name, has a three-letter abbreviation and a one letter code.**

| Amino Acid     | 3 letter code | 1 letter code | Amino Acid    | 3 letter code | 1 letter code |
|----------------|---------------|---------------|---------------|---------------|---------------|
| Glycine        | <b>Gly</b>    | <b>G</b>      | Threonine     | <b>Thr</b>    | <b>T</b>      |
| Alanine        | <b>Ala</b>    | <b>A</b>      | Cysteine      | <b>Cys</b>    | <b>C</b>      |
| Valine         | <b>Val</b>    | <b>V</b>      | Tyrosine      | <b>Tyr</b>    | <b>Y</b>      |
| Leucine        | <b>Leu</b>    | <b>L</b>      | Asparagine    | <b>Asn</b>    | <b>N</b>      |
| Isoleucine     | <b>Ile</b>    | <b>I</b>      | Glutamine     | <b>Gln</b>    | <b>Q</b>      |
| Methionine     | <b>Met</b>    | <b>M</b>      | Aspartic Acid | <b>Asp</b>    | <b>D</b>      |
| Proline        | <b>Pro</b>    | <b>P</b>      | Glutamic Acid | <b>Glu</b>    | <b>E</b>      |
| Phenyl alanine | <b>Phe</b>    | <b>F</b>      | Lysine        | <b>Lys</b>    | <b>K</b>      |
| Tryptophan     | <b>Trp</b>    | <b>W</b>      | Arginine      | <b>Arg</b>    | <b>R</b>      |
| Serine         | <b>Ser</b>    | <b>S</b>      | Histidine     | <b>His</b>    | <b>H</b>      |

# Proteins

## A Polypeptide

### Protein

Phosphofructokinase

Catalase

**Fibrinogen**

Serum albumin

Hemoglobin

**Ovalbumin**

### Mass

345,400

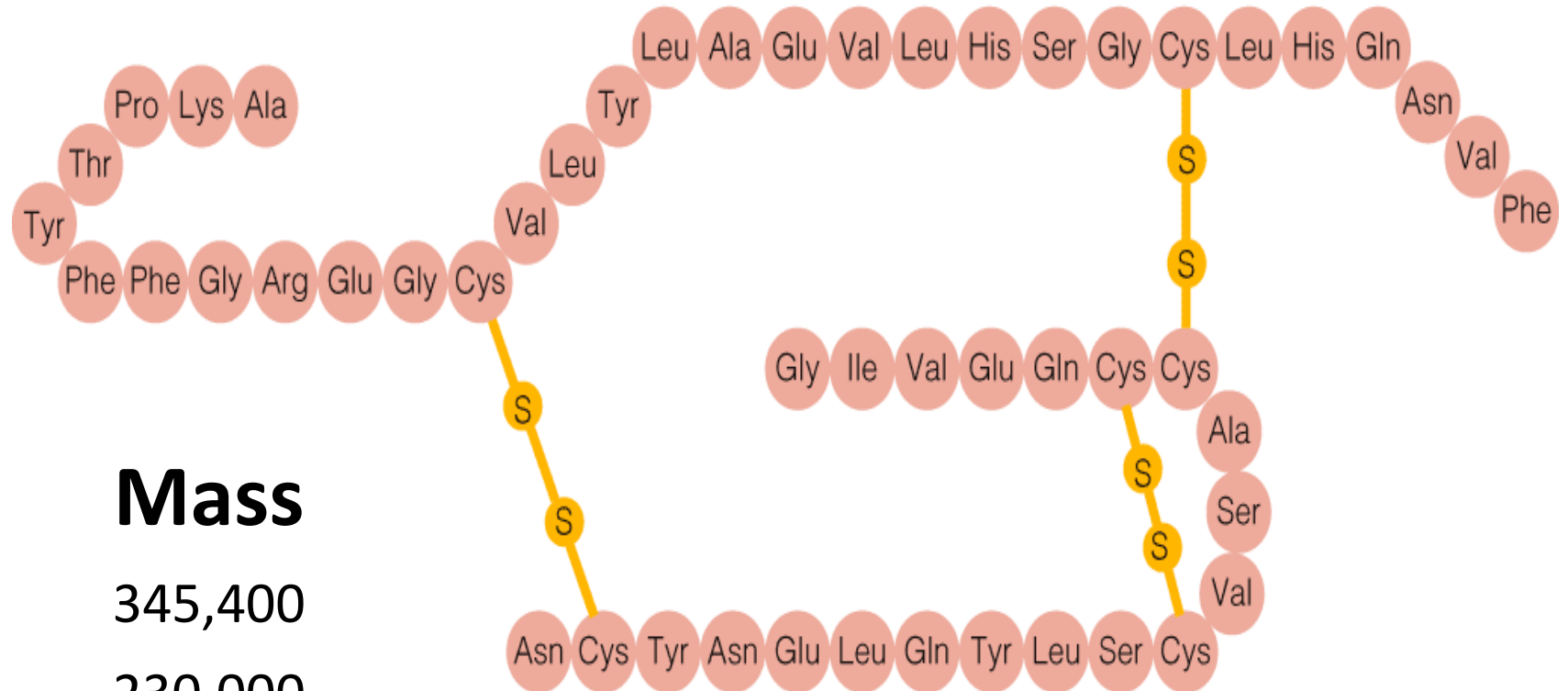
230,000

**390,000**

66,400

64,000

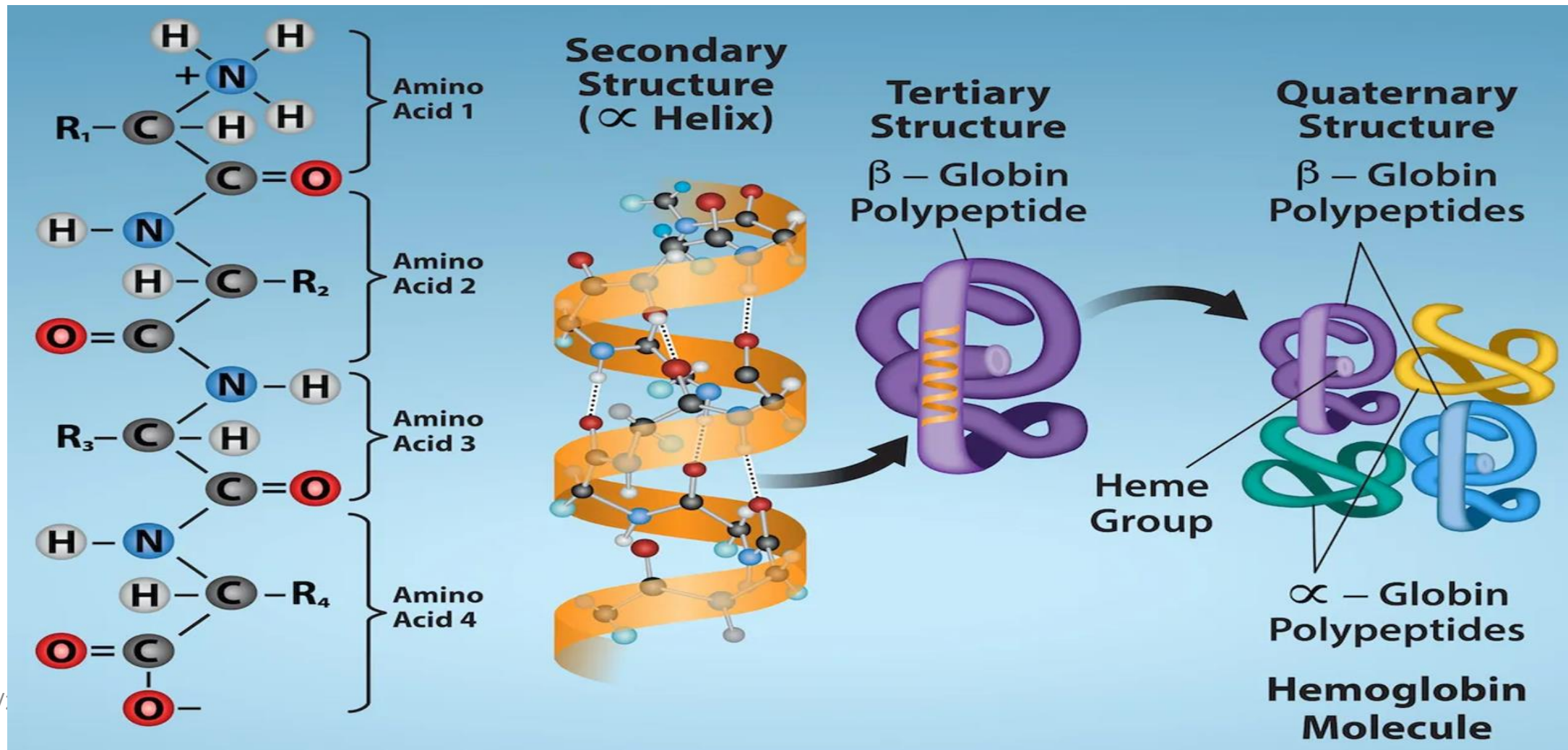
**43,000**





# Protein Structure/Levels of Protein Organization

1. Primary Structure 2. Secondary Structure 3. Tertiary Structure 4. Quaternary structure



# The quaternary structure of proteins

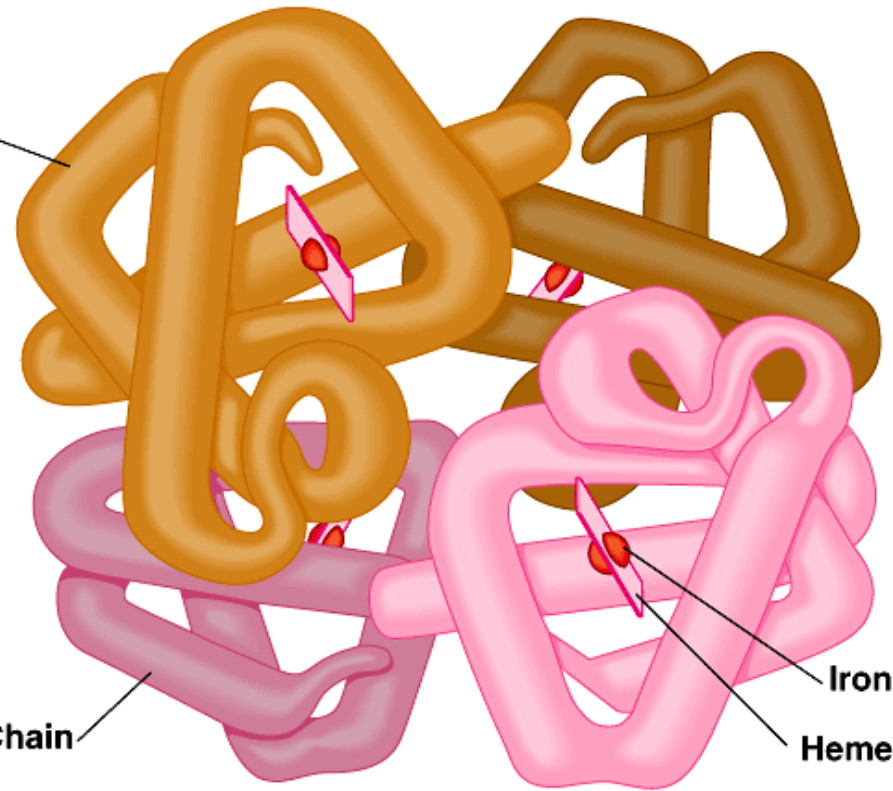


(a) Collagen

Polypeptide chain

$\beta$  Chain

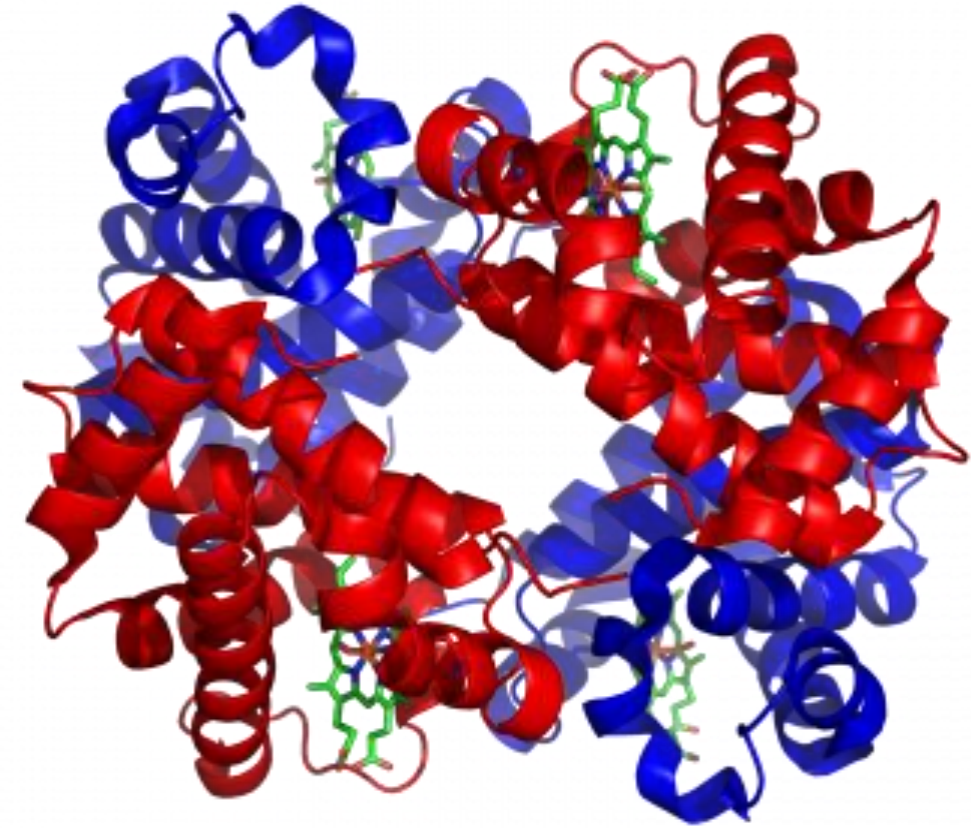
$\alpha$  Chain



Iron

Heme

(b) Hemoglobin

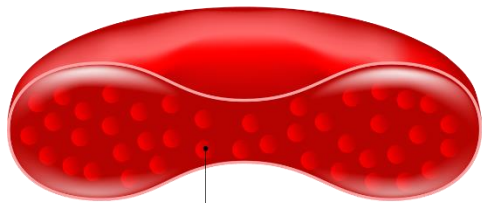
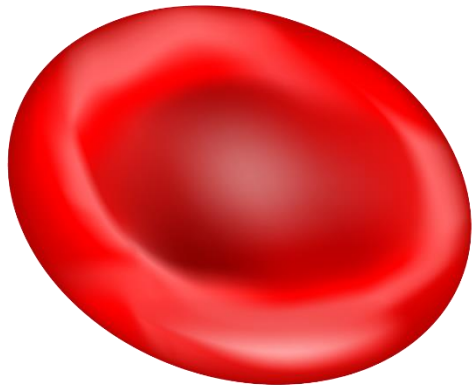


Human Hemoglobin is composed of four polypeptides, two alpha subunits (shown in red) and two beta subunits (shown in blue).



# Protein misfolding and development of sickle cell anemia

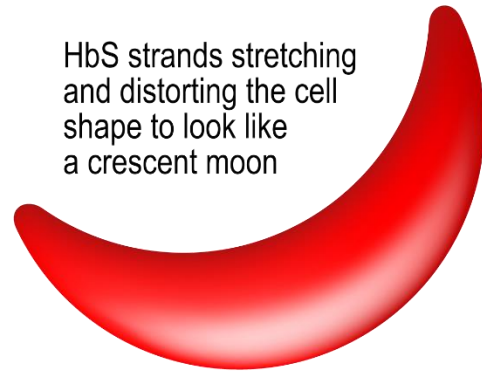
NORMAL RED  
BLOOD CELLS



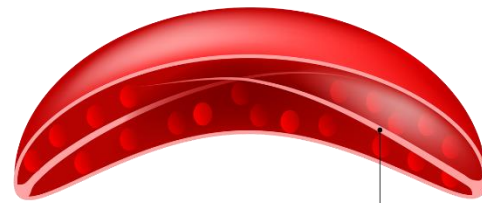
Normal  
haemoglobin

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SICKLE-SHAPED  
RED BLOOD CELLS



HbS strands stretching  
and distorting the cell  
shape to look like  
a crescent moon

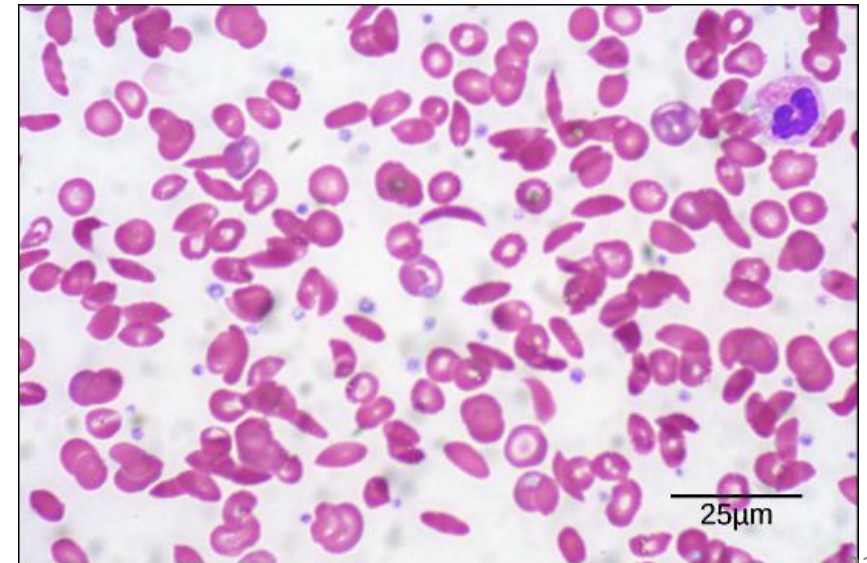


Long polymerized sickle  
haemoglobin (HbS)

Normal RBC



Sickled RBC



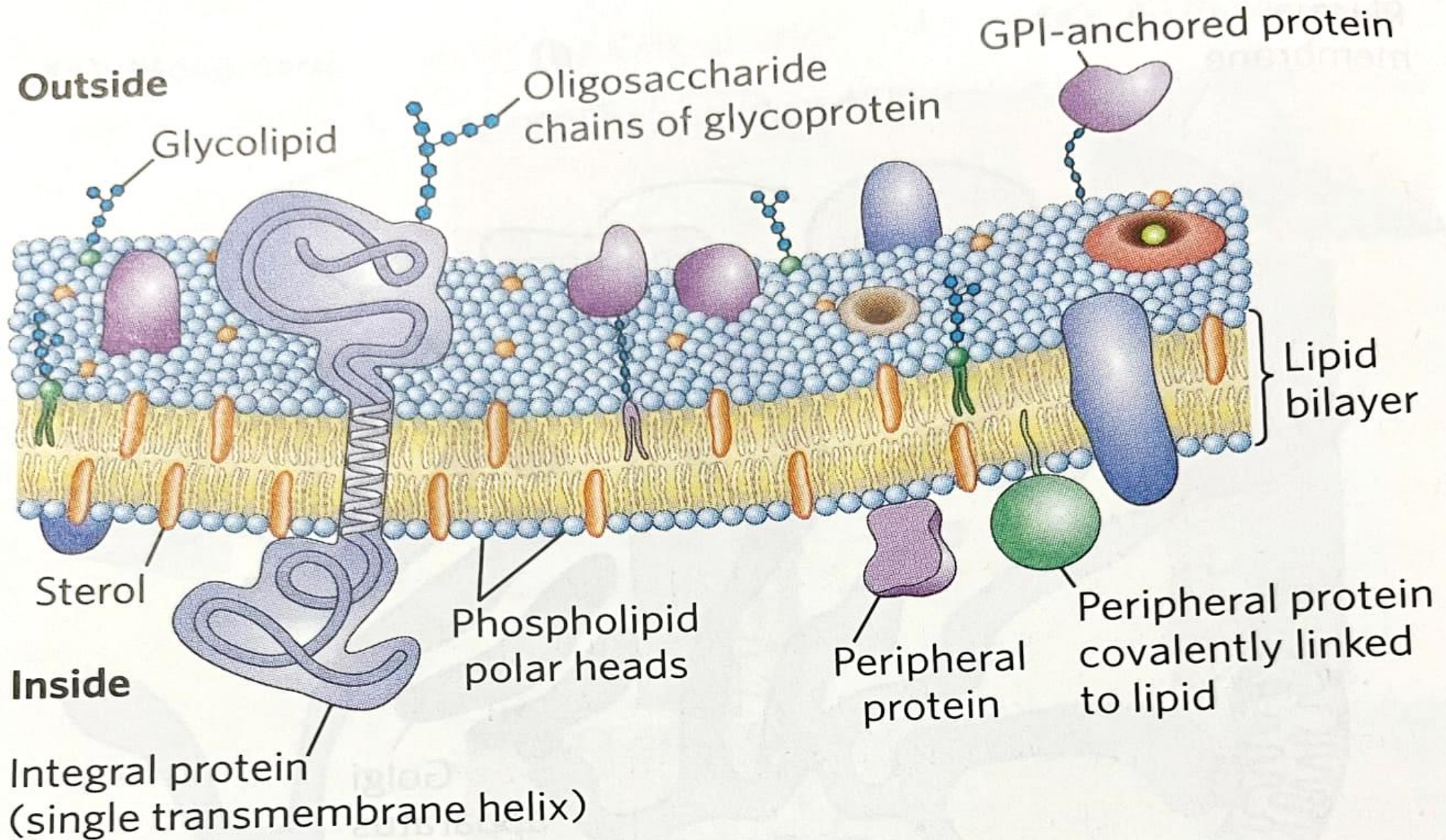


# Transport/LIPIDS



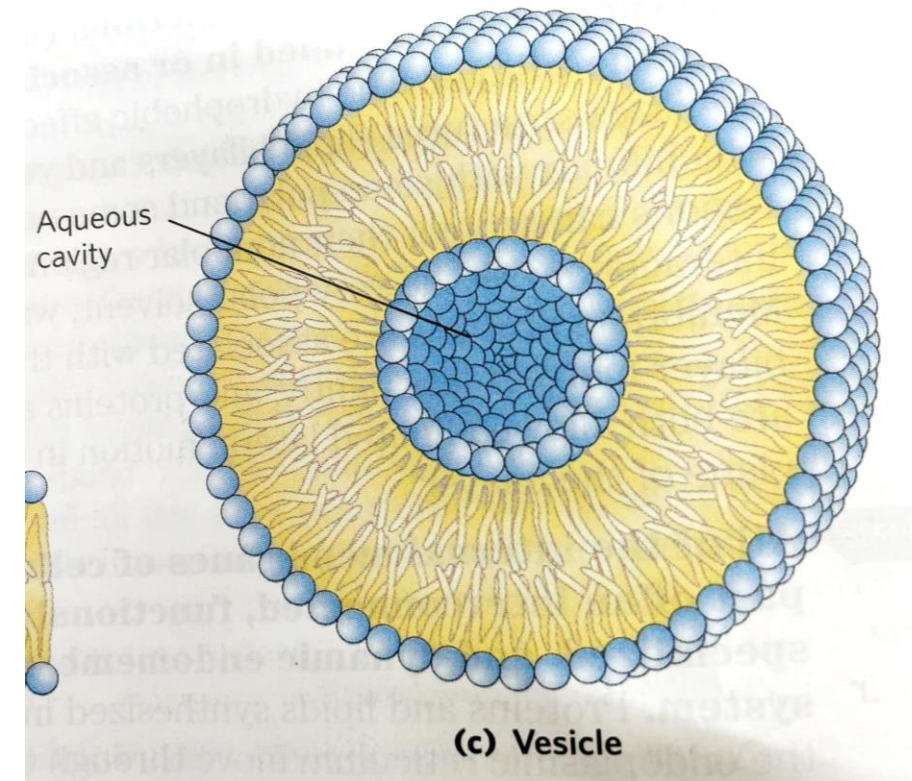
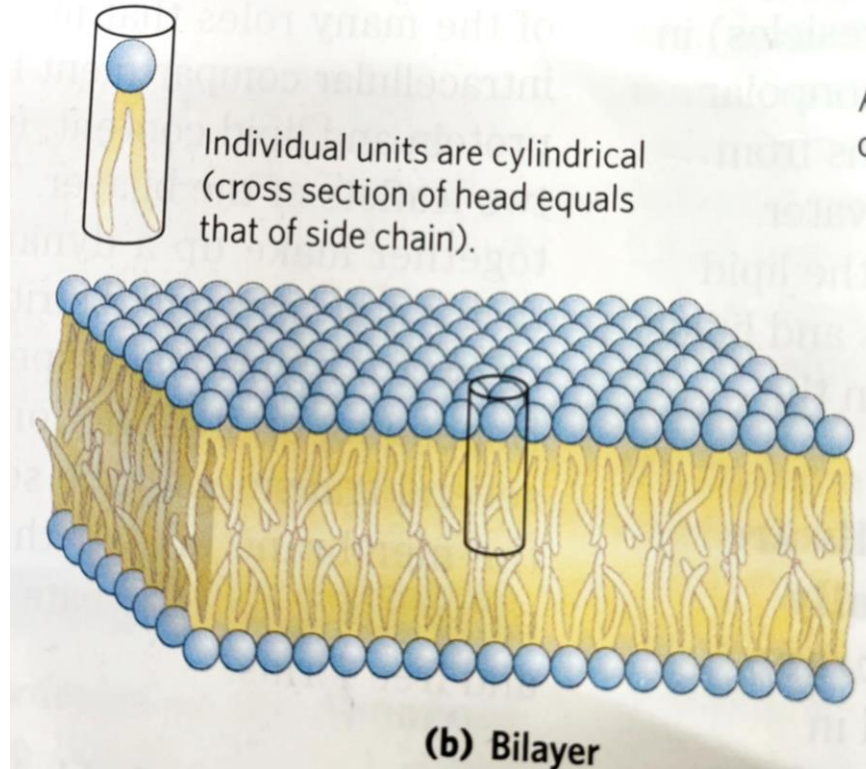
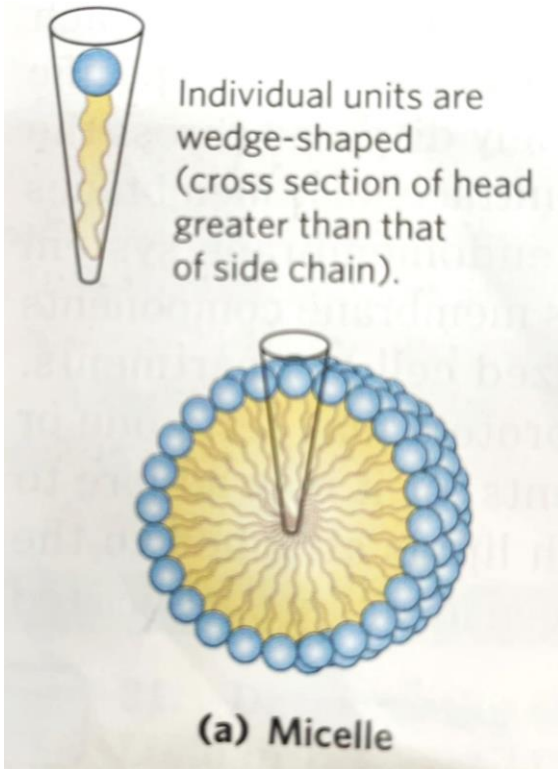


# Cellular Transport



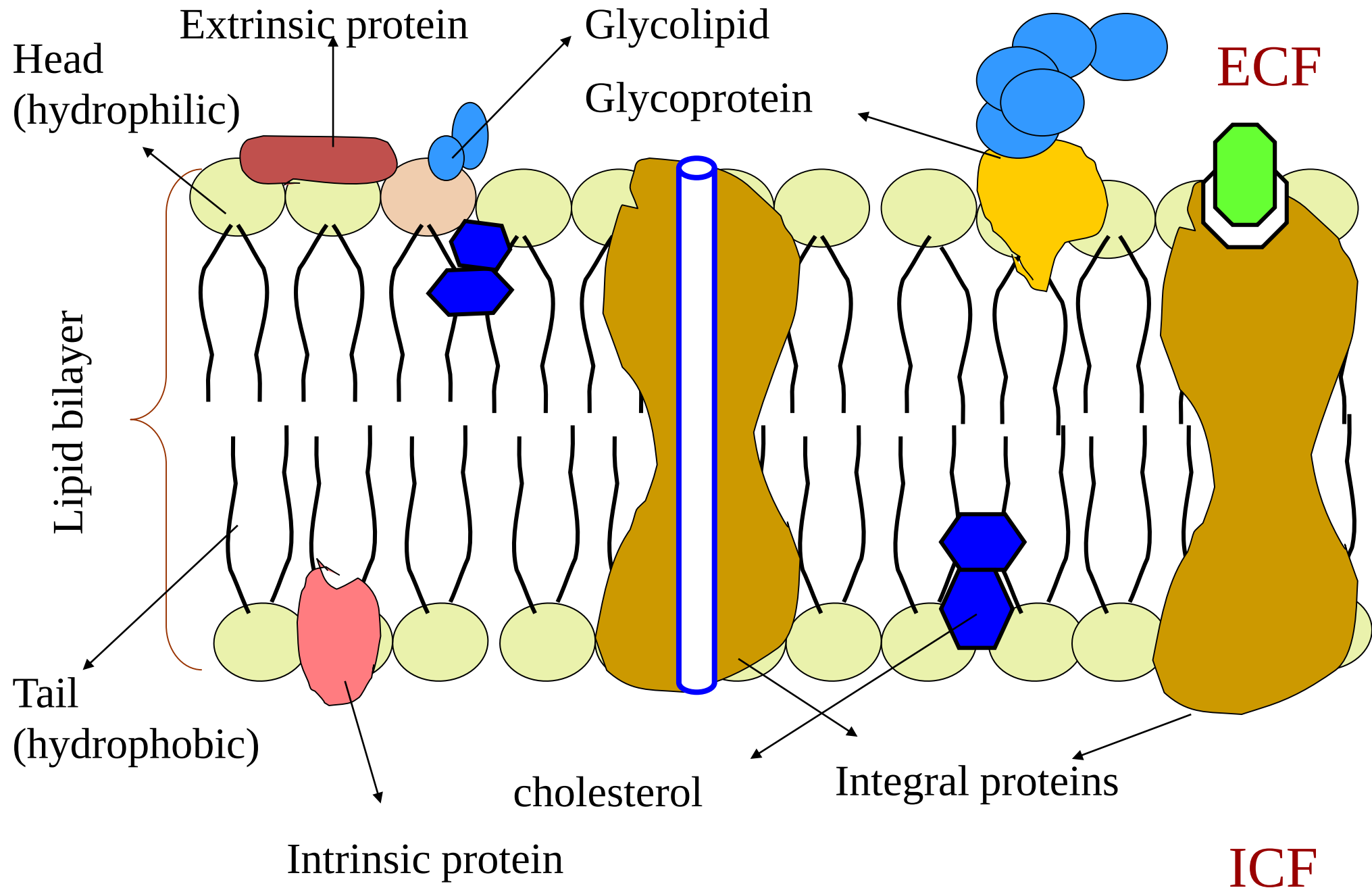


# Amphipathic lipid aggregates that form in water.

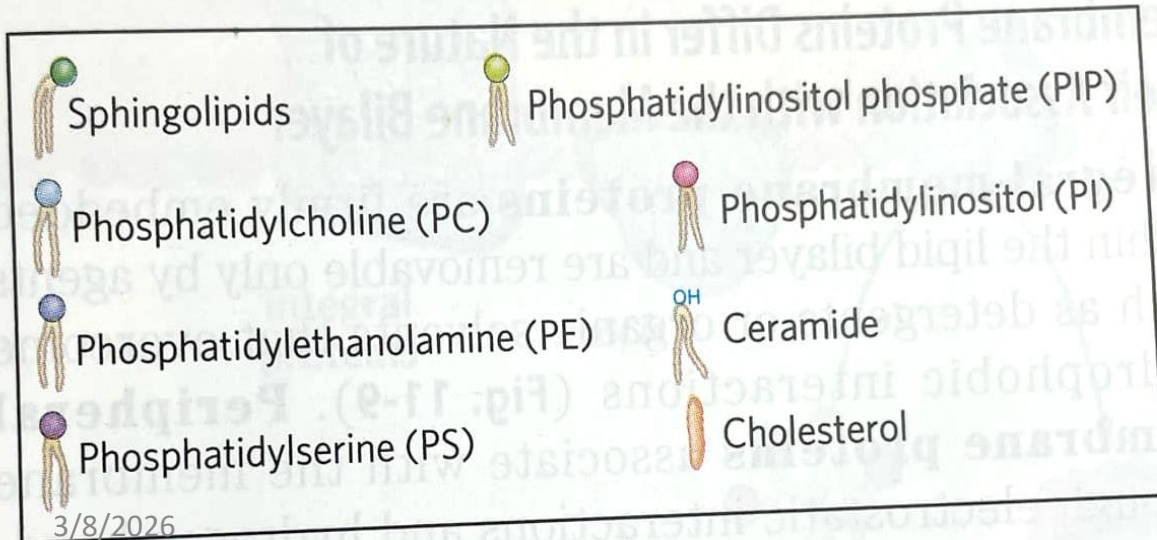
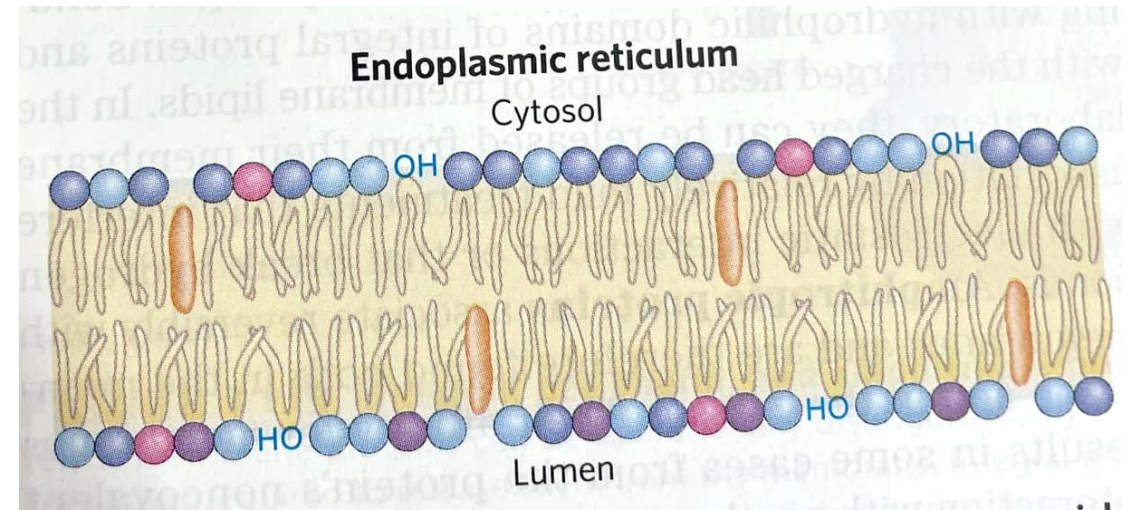
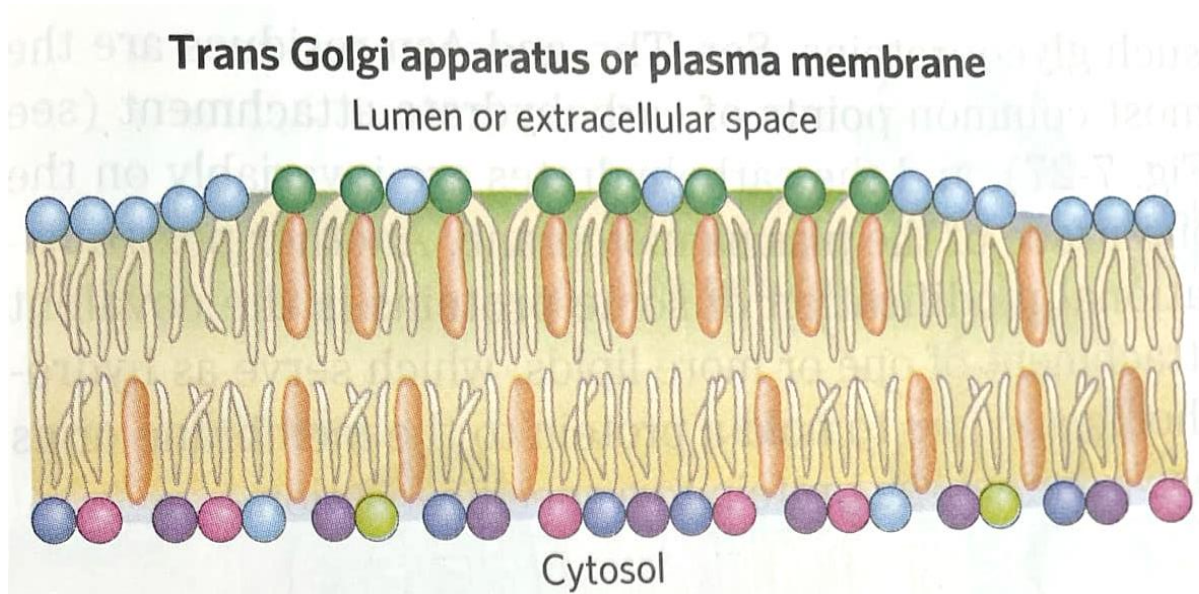


- (a)** In micelles, the hydrophobic chains of the fatty acids are sequestered at the core of the sphere. There is virtually no water in the hydrophobic interior.
- (b)** In an open bilayer, all acyl side chains except those at the edges of the sheet are protected from interaction with water.
- (c)** When a two-dimensional bilayer folds on itself, it forms a closed bilayer, a three-dimensional hollow vesicle (liposome) enclosing an aqueous cavity.





# Membrane compositions:





# Functions of cell membrane

- **Acts as semi permeable barrier –(selective)**
  - Maintains difference in composition of ICF & ECF & fluid in various organelles
  - Protects cell from toxic substances
  - Excretion of waste products
  - Transport of nutrients
  - “Gatekeeper” of the cell—regulates the flow of materials into and out of cell—selectively permeable
- **Receives signals from the outside**
  - Chemical and Electrical signals

# Membrane Transport:

